Homogeneous Functions

LET – Mathematics and Statistics

A function is considered homogeneous, with degree n, if

$$f(kx, ky) = k^n f(x, y)$$

Determine whether the functions below are homogeneous or not. For those that are homogeneous, give the degree of homogeneity.

1. $f(x, y) = 5x + 2y$
2. $f(x, y) = x^2y^3$
3. $f(x, y) = x^2 + y^3$
4. $f(x, y) = 4x^{0.2}y^{0.6}$
5. $f(x, y) = \frac{5x}{2y}$
6. $f(x, y) = \ln\left(\frac{x}{y}\right)$
7. $f(x, y) = \ln x - \ln 2y$
8. $f(x, y) = x^{0.5}e^{\frac{x}{y}}$
9. $f(x, t) = xt - \frac{x^2}{t}$
10. $f(x, t) = xt - \frac{x^3}{t}$